

Intelligent multi-modal systems

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Research in human/computer interaction has mainly focused on natural language, text, speech and vision primarily in isolation. Recently there have been a number of research projects that have concentrated on the integration of such modalities using intelligent reasoners. The rationale is that many inherent ambiguities in single modes of communication can be resolved if extra information is available.

This paper describes an intelligent multi-modal system called the Smart Work Manager. The main characteristics of the Smart Work Manager are that it can process speech, text, face images, gaze information and simulated gestures using the mouse as input modalities, and its output is in the form of speech, text or graphics. The main components of the system are the reasoner, a speech system, a vision system, an integration platform and the application interface. The overall architecture of the system will be described together with the integration platform and the components of the system which include a non-intrusive neural network based gaze-tracking system. The paper concludes with a discussion on the applicability of such systems to intelligent human/computer interaction and lessons learnt in terms of reliability and efficiency.

1. Introduction

The rapid growth in the use (and processing power) of computers both in the home and in the workplace is leading to the situation where the market for 'man/machine traffic' is growing fast. Machine intelligence (MI) will provide the basis for a plethora of new service offerings, not only for the world of business but also for the domestic user of telecommunications. The future will be about added value — not the mere transport of bits [1]; and what could be a better way to add that value than a network which understands and speaks to our customers. The realisation of that vision would truly launch the information age.

The sheer quantity of information available today necessitates co-operation between humans and computers for tasks involving prediction and decision making. Computers can be used as passive tools that respond to precise commands, but this greatly under-utilises their potential. It is important for computers to proactively assist humans, anticipate the effects of their action and learn from users' reactions. Recent advances in artificial intelligence (AI) research and computer hardware have made achievable the vision of a world in which computers work together in teams with humans acting as supervisors who are called upon only in rare situations. In such a scenario each computer possesses a particular skill and, if necessary, its knowledge and expertise is utilised to perform a task. The fundamental research issues are:

- how to best capture the information from the outside world,
- how to extract knowledge from the information captured,
- how to use the knowledge to learn and reason,
- how to integrate robustly all the above components.

For a computer to effectively communicate with humans it must possess communication capabilities that are similar to those of a human, i.e. it must see, hear, understand, question and speak. The key point is that it is the combination of these capabilities that allows humans to communicate effectively. Limiting the communication by blocking some of these modalities clearly reduces the amount of information conveyed. This information, although redundant in some circumstances, can be vital in processing interactions such as: 'Put that there'.

The aim of this project is to provide computers with information about their external world using as many modalities as possible, equip them with human-like reasoning mechanisms, and facilitate adaptation to and learning from the user through observations and task-specific expert knowledge. The influential American Association for Artificial Intelligence has recommended intelligent interfaces to be one of several basic research initiatives with high potential for the next decade [2]. An

intelligent multi-modal system is defined here as a system that combines, reasons with and learns from, information originating from different modes of communication between humans and the computer.

This paper describes the development of an intelligent multi-modal system called the Smart Work Manger (SWM) for a workforce scheduling application. The main components of the system are a reasoner, a speech system, a vision system, an integration platform and an application interface. The main characteristics of the Smart Work Manager are that it can process speech, text, face images, gaze information and simulated gestures using the mouse, while its output is in the form of speech, text or graphics (Fig 1).

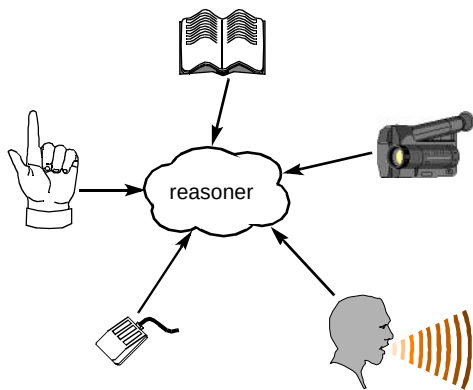


Fig 1 A multi-modal system.

A brief summary of the workforce management system is described in the next section. This is followed by a discussion of the major engineering issues on the intelligent multi-modal interface. The core components such as the reasoner and the neural gaze-tracking system are described in detail, which is followed by a description of the speech systems used. The paper concludes with a discussion on the applicability of such systems to intelligent human/computer

interaction and the lessons learnt in terms of reliability and efficiency.

2. The workforce management system

Intelligent software technology is applicable in situations where the combination of human and current computer technology is either too slow, too expensive or under strain. Examples where machine intelligence will have a beneficial impact on BT operations include communications filtering, telephone receptionist, network management and managers' assistant. The specific application chosen for this project is BT's workforce management system called Work Manager. The key component is the allocation algorithm called Dynamic Scheduler (DS) [3]. The purpose of the DS is to provide BT with the capability:

- to schedule work over a long period of time,
- to repair/optimize schedules,
- to modify the business objectives of the scheduling algorithms,
- to provide statistics from which the schedules can be viewed and their quality assessed.

Data visualisation (DV) is an essential component of the DS. Enormous amounts of information are produced by DS making the assessment of results via traditional management information systems extremely difficult. DV summarises and organises the information produced by DS, facilitating the real-time monitoring and visualisation of work schedules generated.

Both the temporal and spatial dimensions of the information produced are visualised in the Gantt chart and the maps respectively, while statistical information is displayed on various forms and pie charts (see Fig 2). When the system is used, the set of actions/queries during a complete session can be categorised into three classes:

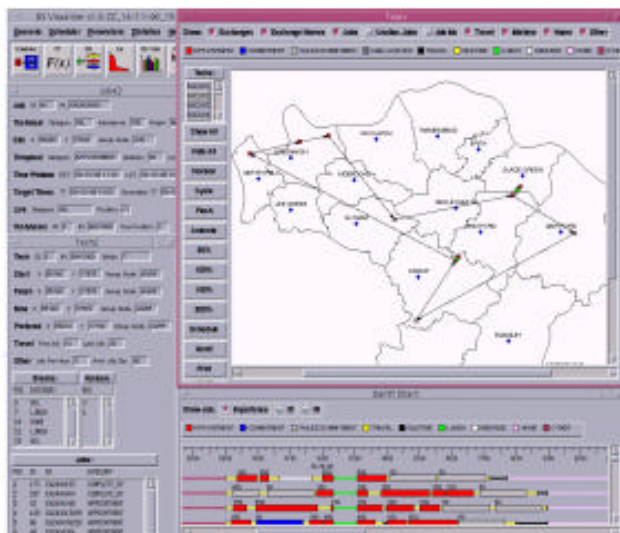


Fig 2 Screen-shots from the Data Visualisation tool.



- queries on scenario and schedule based on textual information display,
- queries on scenario and schedule based on the Gantt chart and the map,
- queries on the scheduling algorithms.

A grammar for these queries is defined as follows:

order/question + object_class + object_identifier + object_attribute

or

order/question + object_attribute + of + object_class + object_identifier

Examples:

{give me} + {technician} + {ab123} + {duration of lunch}

{what is} + {expected completion time} + of + {job} + {10}

{what is} + {duration} + of + {this job}

{show me} + {location in map} + of + {that technician}

3. Engineering issues in building an intelligent multi-modal system

The general aim of this work is to study software architectures for building intelligent multi-modal systems. Before examining such an architecture used for the Smart Work Manager system, some of the general issues pertaining to the engineering of intelligent multi-modal systems are examined below. Intelligent multi-modal systems use a number of input or output modalities to communicate with the user, exhibiting some form of intelligent behaviour in a particular domain. The functional requirements of such systems include the ability to receive and process user input in various forms such as:

- typed text from keyboard,
- hand-written text from a digitiser tablet or light pen,
- mouse movement or clicking,
- speech from a microphone,
- focus of attention of human eye captured by a camera.

The system must also be able to generate output using speech, graphics, and text.

A system which exhibits the above features can be called a multi-modal system. For a multi-modal system to be called intelligent, it is essential that such a system is capable of reasoning in the chosen domain — automating

human tasks, facilitating humans to perform tasks more complex than before and exhibiting a behaviour which can be characterised as intelligent by the users of the system.

Given these requirements for intelligent multi-modal systems, it becomes obvious that such systems, in general, are difficult to develop. A modular approach is therefore necessary for breaking down the required functionality into a number of sub-systems which are easier to develop or for which software solutions already exist.

Another requirement for intelligent multi-modal systems is concurrency. The sub-systems must be running concurrently to process input which may come from more than one input device at the same time. For example, the user may be talking to the machine while at the same time typing text, moving the mouse or gazing at different parts of the screen. The same applies to the output produced by the system. An animation may be displayed while at the same time a speech synthesis program could be running in parallel explaining this animation.

Modular and concurrent systems require some type of communication mechanism which enables information flow between the various sub-systems or modules. Since most intelligent multi-modal systems are going to be experimental at this stage, it is an advantage to be able to easily configure the communications between the different modules. In this way, the system can be quickly adjusted to incorporate new and improved sub-systems.

Given the computational requirements of many of these sub-systems, the distribution of the various modules to run over a network of computers instead of a single machine is also very important. This distribution of the computation implies that the overall system must be able to run on heterogeneous hardware, e.g. Unix workstations, PCs.

To address these requirements, a software platform has been developed which merges existing software from different sources into a single package suitable for building experimental intelligent multi-modal systems.

3.1 Development platform: parallel virtual machine

The core of the platform for building the Smart Work Manager is the Parallel Virtual Machine¹ (PVM) [4] software package. PVM enables a collection of heterogeneous computers to be used as a coherent and flexible concurrent computational resource. The individual computers may be shared on local-memory multiprocessors, vector supercomputers, specialised graphics engines, or scalar workstations, that may be interconnected by a variety of networks, such as Ethernet, FDDI, etc. User programs written in C, C++, FORTRAN, Java and the scripting

¹ Home page: <http://www.epm.ornl.gov/pvm/>

languages, such as Tcl/Tk and Perl, access PVM through library routines.

PVM uses sockets for inter-process communications, but this mechanism is transparent to the user. It essentially provides a computational space where processes (i.e. programs) can run and communicate. In particular, processes can spawn other processes, send messages to other processes, broadcast a message to a group of processes, or kill other processes.

3.2 Programming languages — Tcl/Tk and TkPVM

Tcl² [5] is a simple and easy-to-use scripting language suitable for integrating systems and also for rapid prototyping and development. The core of the language is a small compact command interpreter written in C. The basic interpreter can be extended with extra application-specific commands by incorporating programs either written in C or other computer languages. The internal structure of the interpreter providing PVM, Tk, Tcl and application-specific functionality is shown in Fig 3.

Tk is a widely used extension to Tcl for building graphical user interfaces. Another useful package for Tcl is TkPVM³. TkPVM provides all the necessary functionality for using the PVM functions from within a Tcl script (i.e. Tcl program). Using TkPVM, Tcl scripts can be used as wrappers for applications to connect these applications to PVM. Furthermore, programs already written in Tcl/Tk can immediately make use of PVM and be part of the

distributed system. An example of a distributed system based on PVM is illustrated in Fig 4. This is particularly important in Smart Work Manager as all the systems that have to be integrated are written in different programming languages.

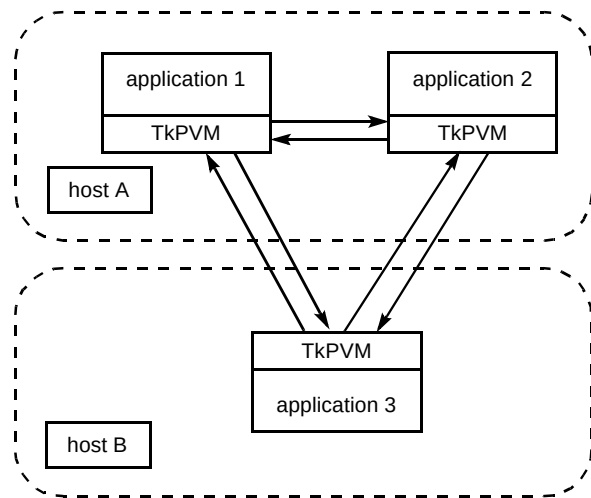


Fig 4 An example of a distributed system using Tcl/Tk and PVM.

3.3 SWM architecture

The main module of the system is a front-end which spawns all the other modules. The user can type questions, correct poorly recognised questions, get answers in text, and see the dialogue history. The overall architecture of SWM with the various modules and the communications between them is given in Fig 5.

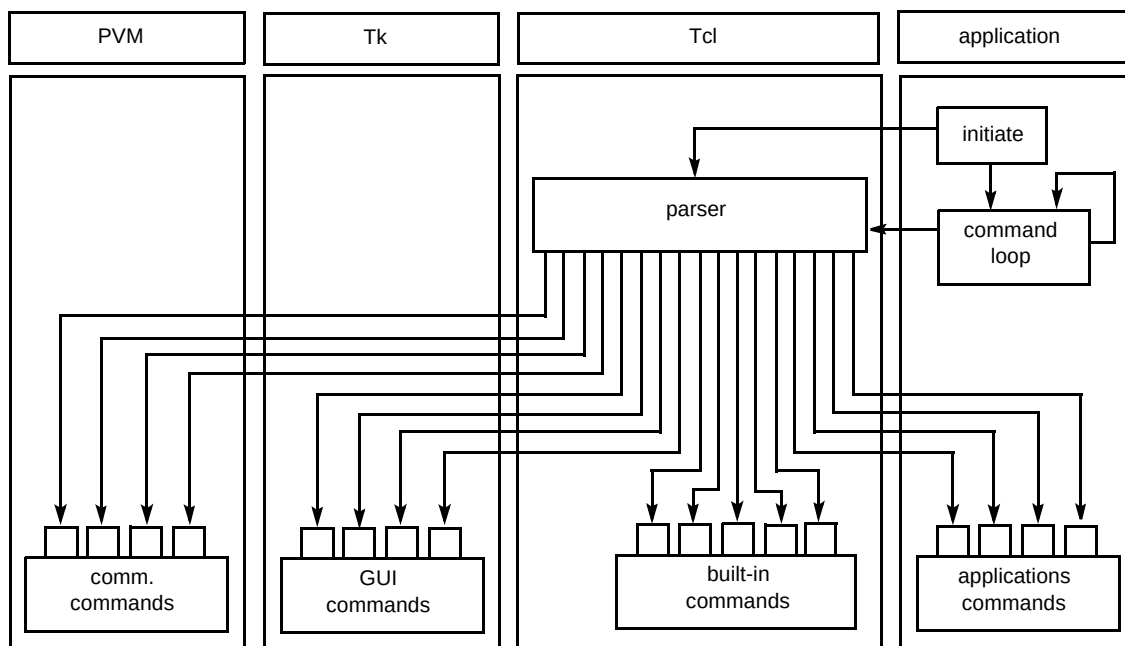


Fig 3 The Tcl/Tk/PVM interpreter.

² Home page: <http://www.sco.com/Technology/tcl/Tcl.html>

³ Home page: <http://www.cogsci.kun.nl/tkpvmm>

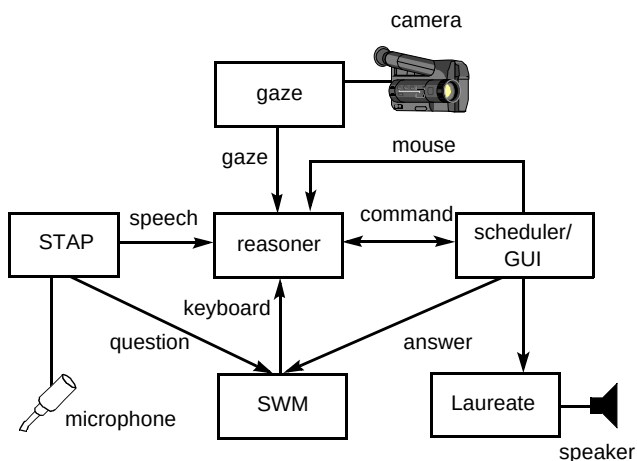


Fig 5 An overview of Smart Work Manager architecture.

4. The reasoner

The main objective of the reasoner can be stated as follows:

Given a sentence which complies with a predefined grammar, convert it to a valid command for the application.

The grammar for the application has been defined in section 2. The main functions of the reasoner are twofold:

- it must be able to handle ambiguities such as: 'give me this of that',
- it must have the capability to deal with conflicting information arriving from various modalities.

The capabilities of the reasoner are to a large extent dependent upon the capabilities provided by the platform on which the reasoner is implemented. The platform used for the reasoner is CLIPS [6], which is a well-known expert system shell with object-oriented, declarative and procedural programming capabilities and the fuzzy logic extension. The reasoner handles ambiguities by using a knowledge base which is being continually updated by the information arriving from various modalities. The structure of the reasoner is shown in Fig 6. There are five modules in the reasoner: fuzzy temporal reasoning — query pre-processing, constraint checking, resolving ambiguities (WIZARD), and post-processing.

The fuzzy temporal reasoning module receives time-stamped events from various modalities and determines the fuzzy temporal relationship between them. It determines to what degree two events have a temporal relationship, such as before, during or overlapping. Using the certainty factors

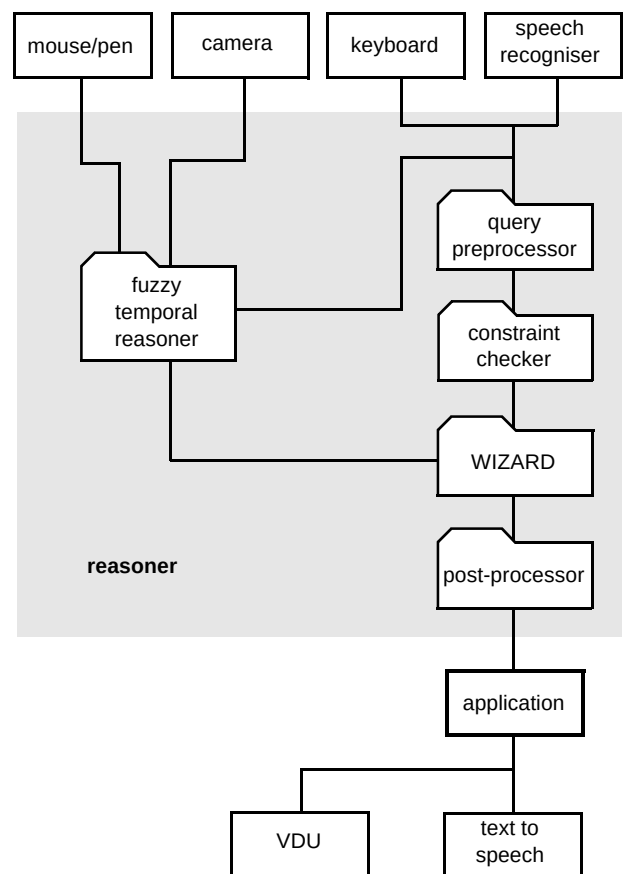


Fig 6 The structure of the reasoner.

(CF) of fuzzyCLIPS [7], the temporal reasoner can answer questions such as:

What is the CF that event 1 took place before event 2?
 What is the CF that event 1 took place just_before event 2?
 What is the CF that event 1 took place during event 2?
 What is the CF that event 1 is overlapping with event 2?

The relationship with the highest CF will be chosen as the most likely relationship between the two events. This relationship can be used later by the reasoner to resolve conflicts between, and checking the dependency of, the modalities.

In the query pre-processing module a sentence in natural language form is converted to a query which conforms to the system's predefined grammar. Redundant words are removed, keywords are placed in the right order and multiple word attributes are converted into single strings.

The constraint-checking module examines the content of the queries. If individual parts of the query do not satisfy predefined constraints then they are replaced by ambiguous terms (e.g. 'this', 'that') to be resolved later, otherwise the query is passed on to the next module.

The WIZARD is at the heart of the reasoner and is the module that resolves ambiguities. The ambiguities in this application take the form of reserved words such as 'this' or 'that', and they refer to objects that the user 'is' or 'has been' talking about, pointing at or looking at. The ambiguities are resolved in a hierarchical manner as shown in Fig 7.

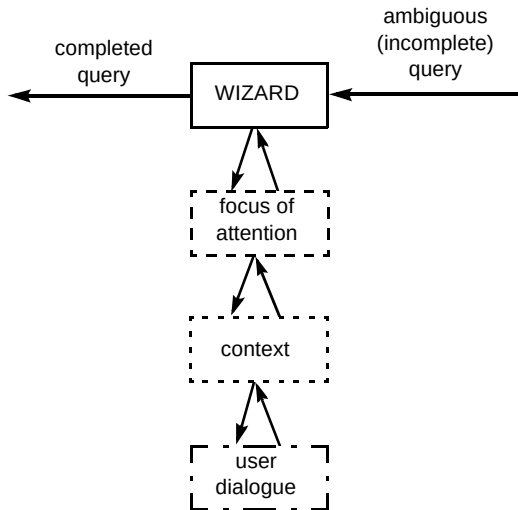


Fig 7 Resolving ambiguities in the reasoner.

The context of the interactions between the user and the system, if it exists, is maintained by the reasoner in the knowledge base. When a new query is initiated by the user, it is checked against the context. If there are ambiguities in the query and the context contains relevant information, the context will be used to create a complete query which will be sent to the application interface for processing.

Another mechanism used to resolve ambiguities is the focus of attention, which is obtained from the user when pointing with the mouse or gazing at an object on the screen. At the same time there could be dialogue with the user through the speech recogniser or the keyboard. CLIPS is mainly used in a declarative mode in this system and therefore all the modules work in parallel. This can cause conflict between information arriving from various modalities. The conflict resolution strategy used in the reasoner is hierarchical as shown in Fig 7, with the focus of attention having the highest priority and the dialogue system the lowest. This means that the dialogue system will act as a safety net for the other modalities if all else fails, or if inconsistent information is received from the modalities.

In cases where text input is required, however, the dialogue system is the only modality which will be called upon. In all other cases the dialogue system will be redundant unless all others fail, in which case a simple dialogue in the form of direct questions or answers will be

initiated by the system. The WIZARD sends the completed queries to the post-processing module.

The post-processing module simply converts the completed queries in a form suitable for the application. This involves simple operations such as formatting the query or extracting key words from it.

Table 1 contains some examples of interactions with the reasoner and how it works.

Table 1 Examples of interaction with the reasoner.

Query	User action	Reasoning process
Show me technician ab 123 job locations on the map.	None.	Complete query, process command and send to application.
Tell me the duration of this job.	Mouse is clicked or eyes are focused on a job.	'This job' is ambiguous, it is resolved using focus. Context is updated.
Show me this of that technician.	No focus. Context is technician 'ab 123 end of day'.	Two ambiguities, context is used to solve ambiguities.
Read me this.	No focus. No context.	Everything is ambiguous, the user is asked to repeat the missing parts. The context is updated.

5. A neural network based real-time non-intrusive gaze-tracking system

Gaze tracking has been a challenging and interesting task studied across several disciplines including machine vision, cognitive science and human/computer interaction [8]. The main applications are in:

- videoconferencing [9] for focusing on interesting objects and transmitting only these images through the communication networks,
- design of new generation of non-command interface [10, 11],
- the study of human vision, cognition, and attentional processes [12].

Traditional methods are usually intrusive [13] and involve a specialised high-speed/high-resolution camera, a controlled lighting source [14], and electronic hardware equipment. Accurate extraction of eye movement information, along with speech, and gestures [15] can

potentially play an essential part in forming a fast and natural interface [16, 17].

The objective of this subsystem is to develop a flexible, cheap, and adequately fast eye tracker using the standard videoconferencing setting on a workstation. The neural network based real-time non-intrusive gaze tracker determines the focus of attention of a user on a computer screen. This task can be viewed as simulating a forward-pass mapping process from the (segmented) eye image space (e.g. the example shown later in Fig 10) to a predefined 2-D co-ordinate space ((x, y) co-ordinates on the screen). In general, the mapping function is nonlinear and highly sensitive to uncertain factors such as change of lighting, head movements and background object movement.

5.1 Methodology and system

The following two assumptions were made:

- the appearance of an eye is sufficient to determine the focus of attention,
- the user's head orientation generally conforms to the line of sight of the eyes.

The former is justified by human/human communications experience, while the latter is introduced to avoid the unnecessary many-to-one mapping situation when a person can look at an object on a screen from various head orientations.

This work was motivated by a recent survey on neural networks [18] which showed successful applications to handle the imprecision in image data and learn the spatial-temporal relations of data sequences for deformable object tracking [19]. Figure 8 shows a schematic diagram of the system which works in an open plan office environment with a simple video camera mounted on the right-hand side of the display screen.

The analogue video signal from a low-cost video camera is captured and digitised using the SunVideo Card, a video capture and compression card for Sun SPARCstations, and the XIL imaging foundation library developed by SunSoft [20]⁴.

For simplicity, only grey-scale images are used in this system⁵. Figure 9 shows a snapshot of the user's face in an open-plan office environment under normal illumination. The objective of the eye image segmentation is to detect the darkest region in the pupil of the eye, and to segment the eye itself from the image (see Fig 10). The current system

uses only the image of the user's right eye. The histogram normalisation improves the contrast between important features (eye socket, pupil, reflection spot) which are used by the neural network during training.

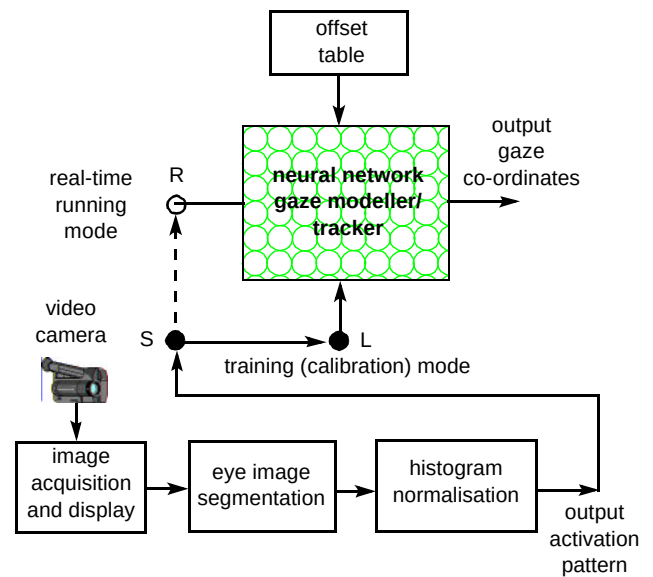


Fig 8 The neural network based gaze-tracking system. The switch S is connected to node L in the training and simulation modes, and to node R in the operation mode (real-time gaze tracking).



Fig 9 User's face in the field of the camera. The rectangle area (100 × 60) in the image defines a search window for eye image segmentation.



Fig 10 Left: an example of the segmented eye image containing the pupil, cornea and the eye socket. Right: eye image after histogram normalisation.

Figure 11 shows the three-layer feedforward neural network. Each of the 600 input units receives a normalised activation value from the segmented 40 × 15 eye image, and the 16 hidden units are divided into two groups of 8 units each. A split output layer of 50 and 40 units represents the

⁴ XIL is a cross-platform C functions library that supports a range of video and imaging requirements.

⁵ Colour images could be used in the future to provide unique features which are not available from grey-scale images [21].

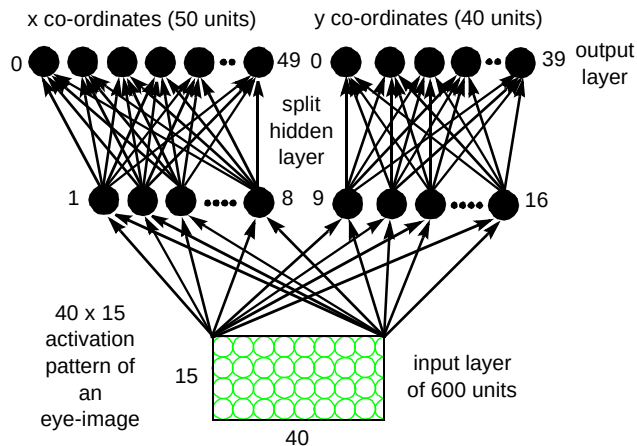


Fig 11 The neural network architecture.

(x,y) co-ordinates of a gaze point on the screen (low resolution grid of 50 by 40). The input units are fully connected to the hidden layer units which function as various feature detectors. All the hidden and output units assume a hyperbolic tangent transfer function (equation 1):

$$f(x) = \frac{1 - e^{-x}}{1 + e^{-x}} \quad \text{..... (1)}$$

with $f(x)$ taking values between -1.0 and 1.0 . A Gaussian-shaped coding method has been adopted for output activations, based on earlier work by Pomerleau [22] on autonomous vehicle guidance and Baluja and Pomerleau [23] on a similar gaze-tracking task.

It is interesting to note that by using the whole eye image directly as the input to the neural network modeller, a global holistic approach to the problem is taken, in contrast to the traditional explicit feature-based approach.

5.2 Experimental studies

For data collection, the user visually tracks a blob cursor which travels along the computer screen in one of the two predefined paths, namely, either the horizontal and vertical movements. Each data example consists of the (x,y) co-ordinates of the travelling cursor and the corresponding eye socket image after segmentation. The backpropagation algorithm (see Bishop [24]) has been modified to train the neural network. The cost function to be minimised is the summed squared error (SSE) which is subject to an evaluation criterion (for stopping purposes) called average grid deviation (AGD). The AGD measures the average difference between the current gaze predictions and the desired gaze positions for the training set excluding a few wildcards due to the user's unexpected eye movements. Although the number of training examples (between 2500 and 4000) is less than the number of free parameters (10 426), the following two training strategies

have made sure that the set of effective parameters were very well tuned, and neither underfitting nor overfitting of the network occurred.

Two phase training method

This first strategy initialises the network with small random weights and consists of a fast search phase where the network is updated every few tens of training examples — this is particularly effective to avoid overfitting and achieves better generalisation performance — followed by a fine-tuning phase where the network weights are updated once, after presenting the whole training set (see Fig 12). Repeated training trials using different initial random seeds and on data examples from several users have demonstrated consistently good performance of the training strategy.

Cross-validation training

This strategy randomly divides the original data set into a training set and a validation set. The validation set is used to monitor the progress of the learning process in order to prevent over-fitting. The learning process stops when the validation error starts to increase or saturates after a general downwards trend.

5.3 A real-time gaze-tracking system

After the training (calibration) process has determined an optimal set of weights, the gaze tracker is ready to run. Captured eye images are fed into the network which outputs the (x,y) co-ordinates of the corresponding gaze point on the screen. The system works at about 20 Hz in its stand-alone mode on a SunUltra-1 Workstation. The average prediction accuracy on a separately collected test data set is about 1.5 degrees, which corresponds to an error of around 0.5 inch on the computer screen.

6. Speech in multi-modal systems

The primary function of a speech recogniser in the multi-modal system is to recognise spoken commands in the form of complete sentences which can then be passed on to the reasoning module for further processing. In order to provide a speech interface as close to human interaction as possible, the commands have to be issued in a natural way as if the user is talking to another person. This requirement effectively precludes the possibility of employing keyword speech recognisers as the users are normally required to speak in a robot-like manner; speaking a word at a time. Another problem with keyword recognisers is the need for an interpreter that can understand the keywords and translate the string of keywords into a meaningful sentence. Error detection, context checking and discourse management would also be complicated.

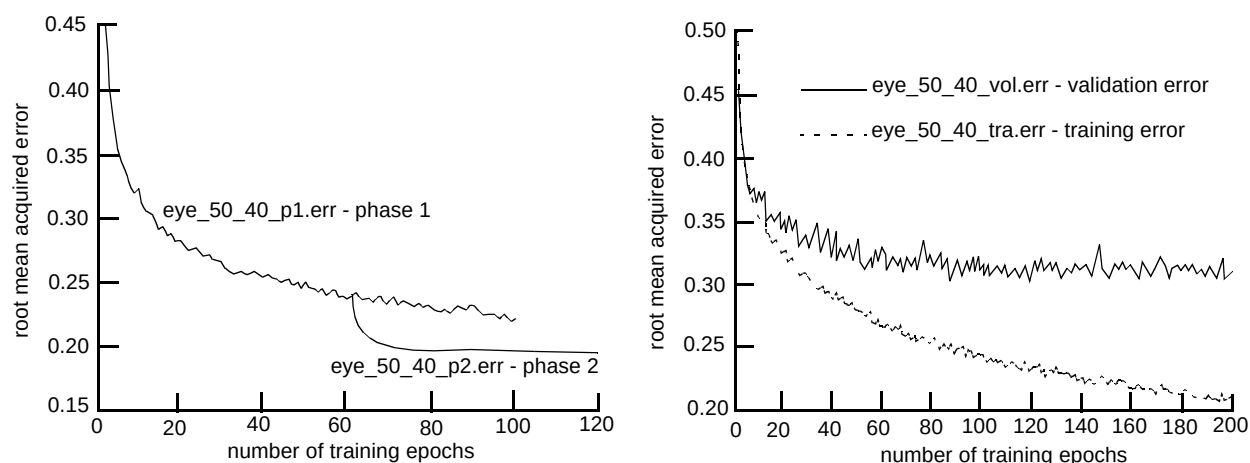


Fig 12 Training errors against number of training epochs (left — two-phase training method; right — cross-validation training).

A speech synthesiser is limited to provide feedback to the user regarding the result of the query in this current application. Due to the dynamic nature of the answers to the queries, pre-recorded messages are not useful. A speech synthesiser can further be developed for better communication between the computer and the user especially when the system needs to engage in a dialogue with the user.

The speech recogniser used in the multi-modal system is StapRec from BT's speech technology unit. StapRec is designed for use over telephone lines which have a limited audio bandwidth. It is speaker independent and works on connected word recognition. Laureate [25] is used in SWM which is the text-to-speech synthesiser developed at BT Laboratories. It is a platform-independent software with the aim of providing natural speech output. The provision of application programming interfaces in these packages allows development of applications without detailed knowledge of speech recognition and synthesis.

StapRec is run as two processes on the PVM platform — a speech recogniser and a voice activity detector. Speech data is passed to StapRec server once voice activity is detected. When a recognition result is available, it is passed to the reasoner for further processing before the actual query is performed.

Laureate is being run in shell scripts under PVM. Initialisation is performed once. Whenever speech synthesis is required, the text is processed to produce an audio file which will then be played on the audio channel by a standard utility program.

7. Related work

Research in human/computer interaction has mainly focused on natural language, text, speech and vision primarily in isolation. Recently there have been a number of research projects that have concentrated on the integration of such modalities using intelligent reasoners. The rationale is that many inherent ambiguities in single modes of com-

munication can be resolved if extra information is available. A rich source of information for recent work in this area is the book by Sullivan and Tyler [26], who review, among other projects, CUBRICON from Calspan-UB Research Centre, XTRA from DFKI (German Research Centre for AI), and the SRI system from SRI International.

The CUBRICON system is able to use a simultaneous pointing reference and natural language reference to disambiguate one another when appropriate. It also automatically composes and generates relevant output to the user in co-ordinated multimedia. The system combines natural language, text commands, speech and simulated gestures such as pointing with a mouse. The application area is military-based maps.

The XTRA system is an intelligent multi-modal interface to expert systems that combines natural language, graphics and pointing. It acts as an intelligent agent performing a translation between the user and the expert system. The most interesting aspect of this project is how free-form pointing gestures such as pointing with fingers at a distance from the screen has been integrated with graphics and natural language to allow a more natural way of communication between the user and the expert system.

The SRI system combines natural language/speech with pen gestures such as circles and arrows to provide map-based tourist information about San Francisco.

At the heart of all the above systems is a reasoner that combines the general and task-specific knowledge in its knowledge base with often vague or incomplete user requests in order to provide a complete query to the application.

The Smart Work Manager described in this paper utilises most of the functionality of the above systems. However, it is viewed as an initial attempt to understand the main issues involved in building a platform for such

systems. Emphasis is put on the fusion of possibly redundant information for various channels.

8. Conclusions

This paper has described an intelligent multi-modal user interface that delivers enhanced communications channels between human and computer in the context of BT's workforce management system. The contribution of this work is on the combination of multiple input modalities in order to support natural human/computer interactions. It has been demonstrated that soft computing techniques are useful tools for such applications. Specifically, the fuzzy temporal reasoner collects and reasons with redundant sources of information to infer the user's intention. Ambiguities are resolved by the reasoner with its context-sensitive fuzzy matching. This allows the user to be vague when issuing commands within the same context of current interaction, as happens in human interactions.

Another major contribution is the real-time non-intrusive gaze-tracking system based on powerful neural network modelling techniques. In contrast with other gaze-tracking systems, the current system works in an office environment under normal illumination, without resorting to any specialised hardware. The Smart Work Manager has also integrated speech input/output modules which has greatly enhanced the naturalness of the user interface.

System integration was a big engineering challenge of this project as almost all the modules in the system are written in different programming languages. By encapsulating individual modules within a generic wrapper written in a scripting language, the subsystems virtually become homogenous entities. Robustness of the system is provided by a flexible PVM platform which allows seamless and efficient communication between subsystems running across a host of heterogeneous hardware. This forms a firm basis for future experiments when new or enhanced modules become available. Moreover, the modular design of the Smart Work Manager means individual components can easily be reused in other systems that require an intelligent multi-modal user interface.

One constraint of the current system is a restricted command set which is shared between the speech recognition system, the reasoner and the Work Manager system. An adaptive interface which supports automatic generation of grammar on new commands will certainly enhance the flexibility of the system. Part of this is also achieved by the speaker-independent speech recogniser. Constraints of the gaze tracking are relatively long training time, not coping well with the change of camera orientation, and apparent head movements. These are expected to be dealt with in future research work.

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